

Technical Challenge – Sourcing Analysis with AI

1. Executive Summary

Objective

The purpose of this analysis is to identify the sourcing strategies, candidate characteristics, and recruitment patterns that are most strongly associated with successful hiring outcomes.

By analyzing the recruitment funnel and candidate progression data, the goal is to generate actionable recommendations that help recruiters prioritize efforts and improve conversion rates.

Main Findings

- GitHub and Inbound sourcing channels showed the highest hiring conversion rates.
 - Senior and Staff candidates demonstrated stronger progression throughout the recruitment funnel.
 - Assessment scores, particularly manager evaluations, were strong indicators of hiring success.
 - Candidate response time was associated with conversion likelihood.
 - Significant differences were observed between recruiter performance metrics.
-

2. Analytical Approach

Data Understanding

The dataset contains candidate sourcing and recruitment process information, including:

- Candidate profile attributes
- Sourcing channels
- Recruiter ownership
- Funnel progression stages
- Assessment scores
- Hiring outcomes

Analysis Framework

The analysis was conducted in four stages:

1. Exploratory Data Analysis (EDA)
2. Funnel Conversion Analysis

3. Driver Analysis
4. Recommendation Generation

The objective was to identify which factors influence candidate advancement and final hiring outcomes.

3. Key Decisions Made

Funnel Definition

The recruitment funnel was defined using the following stages:

- Sourced
- Response Received
- Screening Passed
- Interview Passed
- Offer Sent
- Hired

This structure allows clear visibility into conversion rates and drop-off points.

Conversion Metrics

Conversion rates were calculated both overall and segmented by:

- Source Channel
- Recruiter
- Seniority
- Candidate Assessment Scores

Candidate Signals

To identify hiring predictors, candidate groups were compared across:

- Technical Assessment Scores
 - Behavioral Scores
 - Manager Evaluation Scores
 - Years of Experience
 - Seniority Level
-

4. Use of AI

AI was incorporated to enhance insight generation and recruiter decision support.

AI Applications

Insight Exploration

Generative AI was used to support hypothesis generation and identify potential relationships between sourcing channels, candidate characteristics, and hiring outcomes.

Recruiter Copilot Concept

A simple AI-assisted recommendation workflow was designed to translate analytical findings into recruiter-friendly guidance.

Example output:

"High Priority Candidate"

Reasons:

- Strong technical assessment
- High manager evaluation
- Positive sourcing channel performance

Recommended Action:

Prioritize engagement and accelerate process progression.

This demonstrates how AI can transform data insights into practical recruiting actions.

5. Key Insights

Recruitment Funnel

The largest candidate drop-off occurred between the response and screening stages, indicating an opportunity to improve candidate qualification and outreach targeting.

Sourcing Performance

GitHub and Inbound channels generated the strongest hiring conversion rates, suggesting these channels produce higher-quality candidate pipelines.

Recruiter Performance

Meaningful differences were observed across recruiters, indicating opportunities to identify and replicate best practices.

Candidate Success Indicators

The strongest predictors of hiring success were:

- Higher manager evaluation scores
- Higher technical assessment scores
- Senior and Staff-level profiles

Candidate Engagement

Candidates who responded within the first week maintained meaningful hiring probability, supporting continued recruiter follow-up efforts during this period.

6. Recommendations

Top of funnel

- Increase investment in sourcing channels with higher hiring conversion rates.
- Establish recruiter best-practice sharing based on conversion performance.

Middle of funnel

- Use candidate assessment scores as prioritization signals during pipeline management.
- Maintain structured follow-up efforts for up to seven days after initial outreach.

End of funnel

- Implement AI-assisted candidate prioritization to support recruiter decision-making and improve efficiency.
-

Conclusion

The analysis identified clear relationships between sourcing strategies, candidate characteristics, recruiter performance, and hiring outcomes.

Combining data-driven recruiting practices with AI-assisted recommendations can improve recruiter efficiency, increase conversion rates, and support more effective talent acquisition decisions.